

Informatics

Numerical Methods

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Learning Objectives

- Show some solutions of mathematical problems through computation, instead of analytics
- see the `numerical_methods` folder on GitHub

Finding the root of a function

find x such that $f(x) = 0$

Finding the root with the Newton method

- let $f(x)$ have the derivative $f'(x)$
- based on the Taylor approximation
- convergence is not guaranteed in general

$$x_0 = \text{initial guess}$$

$$x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$$

$$f(x_n) \approx f(x_{n-1}) + (x_n - x_{n-1})f'(x_{n-1})$$

see example `newton.R`

Zero of a function with the *bisection* algorithm

- start with two values such that $f(x_1) * f(x_2) < 0$
 - the sign is opposite
- if f is continuous the zero will lie somewhere between x_1 and x_2
- choose the midpoint x between x_1 and x_2
- choose the next interval accordingly to the sign of $f(x)$
- stop looping when a pre-set precision is met

see example `bisect.R`

Finding the minimum of a function

- analytic solutions are based on the derivative of the function
- it happens that a function is not differentiable in the interval of interest
- the minimum can be found by search
- the quality of the solution is related to
 - precision
 - convergence speed
- necessary a strategy to choose the test points

Golden Section search method

Braun, Murdoch, "Statistical programming with R", Cambridge

- valid for a function that has a single minimum on a given interval $[a,b]$
- iterative solution
 - based on a loop that ends when a given stop condition is met
- 1. start with the interval that is known to contain the minimizer
- 2. repeatedly shrink it, finding smaller and smaller intervals $[a',b']$ which contain the minimizer
- 3. stop when $b'-a'$ is small enough, i.e. when the interval length is less than a pre-set *tolerance*
- 4. use as minimizer the midpoint of the last interval
- 5. the maximum error is $(b'-a')/2$

Golden Section Search (ii)

- Let's name x_1 and x_2 the points where we will test the function
- in step 2 we choose $x_1 < x_2$
 - we will see how in a short while
- if $f(x_1) > f(x_2)$ the minimizer must lie to the right of x_1
 - new interval $[a', b'] = [x_1, b]$
- if $f(x_1) < f(x_2)$ the minimizer must lie to the left of x_2
 - new interval $[a', b'] = [a, x_2]$
- if they are equal simply choose always one side
- choose new values x_1 and x_2 and compute the function in the chosen points, until the stop condition is met

The Golden Ratio

- interesting algebraic properties

$$\phi = (\sqrt{5} + 1)/2$$

$$1/\phi = \phi - 1$$

$$1/\phi^2 = 1 - 1/\phi$$

Golden ratio $\cong 0.618$
therefore $x_1 < x_2$

$$x_1 = b - (b - a)/\phi$$

$$x_2 = a + (b - a)/\phi$$

Golden Ratio Search

- after one iteration it is possible that we throw away a and replace it with $a' = x_1$
- new value to use as $x_1 \rightarrow$
- i.e., we can re-use a point already used, without need of a new calculation

$$\begin{aligned}x'_1 &= b - (b - a')/\phi \\&= b - (b - x_1)/\phi \\&= b - (b - a)/\phi^2 \\&= a + (b - a)/\phi \\&= x_2\end{aligned}$$

see example `golden_search.R`